METRIC
MIL-T-24735(NAVY)
31 August 1989

MILITARY SPECIFICATION

TRANSMITTERS, LIGHT SIGNAL, ANALOG, FIBER OPTIC, SHIPBOARD (METRIC), GENERAL SPECIFICATION FOR

This specification is approved for use by the Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for fiber optic analog transmitters for use in Naval shipboard systems. The transmitter modulates an optical carrier with an electrical input analog signal and transmits the signal through the optical output interface. The transmitter does not alter the modulation format.

1.2 Classification. Transmitters are classified as specified in 1.2.1 through 1.2.4 (see 3.1).

1.2.1 Type. The type designation defines the type of light source.

(a) Type 1 - Light emitting diode (LED).
(b) Type 2 - Laser diode (LD).

1.2.2 Wavelength class. The wavelength class indicates the nominal wavelength of output optical signals.

(a) Class A - 1.31 micrometers (μm).
(b) Class B - 1.55 μm.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6025

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1.2.3 Frequency (see 6.6.6). The maximum frequency of operation of the transmitter falls within one of the following frequency ranges.

(a) Range 1 - 0 to 20 megahertz (MHz).
(b) Range 2 - 20 to 500 MHz.
(c) Range 3 - 500 MHz to 2 gigahertz (GHz).
(d) Range 4 - Greater than 2 GHz.

1.2.4 Style. The style designation defines the style of packaging.

(a) Style A - Dual inline package using multimode pigtail.
(b) Style B - Dual inline package using single-mode pigtail.
(c) Style C - Dual inline package, multimode, connectorized.
(d) Style D - Dual inline package, single-mode, connectorized.
(e) Style E - Surface mount package using multimode pigtail.
(f) Style F - Surface mount package using single-mode pigtail.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-T-24735/1 - Transmitter, Analog, Fiber Optic, Shipboard, 0.5 to 60 MHz (0.5 dB Passband) (Metric).
MIL-R-24737 - Receivers, Light Signal, Analog, Fiber Optic, Shipboard (Metric), General Specification for.
MIL-R-24737/1 - Receiver, Light Signal, Analog, Fiber Optic, Shipboard, 0.5 to 60 MHz (0.5 dB Passband) (Metric).
MIL-M-38510 - Microcircuits, General Specification for.
MIL-M-55565 - Microcircuits, Packaging of.
MIL-C-85045 - Cable, Fiber Optic, Shipboard (Metric) General Specification for.
MIL-C-85045/14 - Cable, Fiber Optic, Fiber Optic Cable Configuration Type B (Pigtail), Optical Fiber Type MM (Graded Index, Glass Core and Glass Cladding, Multimode) (Metric).

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STANDARDS

MILITARY

MIL-STD-202	- Test Methods for Electronic and Electrical Component Parts.
MIL-STD-454	- Standard General Requirements for Electronic Equipment.
MIL-STD-461	- Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-462	- Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-810	- Environmental Test Methods and Engineering Guidelines.
MIL-STD-883	- Test Methods and Procedures for Microelectronics.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

ELECTRONIC INDUSTRIES ASSOCIATION (EIA)

455-6	- Cable Retention Test Procedure for Fiber Optic Cable Interconnecting Devices. (DoD adopted)
455-36	- Twist Test for Fiber Optic Connecting Devices. (DoD adopted)

(Application for copies should be addressed to Electronic Industries Association, 2001 Eye Street, NW, Washington, DC 20006.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern.

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3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.4) in accordance with 4.4.

3.3 Reliability. Transmitter reliability shall be expressed as a mean time between failure (MTBF), in hours of operation, and shall be as specified (see 3.1). Reliability shall be determined by procedures specified in 4.6.1 (see 6.2). Transmitter failure criteria shall be as specified (see 3.1).

3.3.1 Burn-in (see 4.3.2). Burn-in requirements of MIL-STD-883, method 1015 shall be met.

3.4 Materials. Materials shall not have adverse effects upon operational or maintenance personnel under all operational and environmental conditions, nor cause degradation of equipment performance (see 6.3).

3.4.1 Recovered materials. Unless otherwise specified, all equipment, material, and articles incorporated in the products covered by this specification shall be new and may be fabricated using raw materials produced from recovered bulk materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of partially processed, assembled, or rebuilt products is allowed under this specification.

3.5 Design and construction. The transmitter shall be of the design and construction as specified herein.

3.5.1 Fiber optic pigtail. Transmitter styles A, B, E, and F shall be supplied with a fiber or cable pigtail (see 6.6.10 and 6.6.11) as specified (see 3.1). One end of the pigtail shall be permanently mounted in the transmitter case, and the mounting construction may provide strain relief for the pigtail in order to meet the requirements of 3.11. The pigtail length shall be not less than 1 meter. No metal components shall be used within the pigtail, except for mounting to the transmitter case or termination connector.

3.5.2 Fiber optic connector. Transmitter styles C and D shall be supplied with a fiber optic connector as specified (see 3.1).

3.5.3 Optical source. The transmitter shall use an LED or an LD as specified (see 1.2.1).

3.5.4 Case (see 4.6.2.1). The transmitter shall be hermetically sealed. Dimensions, pinouts, and pinout functions shall be as specified (see 3.1). The number of pins on each transmitter shall be one of the following: 14, 16, 18, 20, 24, or as specified (see 3.1). The transmitter case may be electrically grounded, connected to a voltage pin, or electrically isolated (that is, floating) as specified (see 3.1).

3.5.5 Terminals. The terminals shall meet the requirements as specified in 3.5.5.1 and 3.5.5.2.

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3.5.5.1 Terminal strength (see 4.6.2.2). The electrical terminals shall reveal no evidence of breakage, loosening, or relative motion between the electrical terminals and the case of the transmitter. The terminals shall meet the requirements as specified (see 3.1).

3.5.5.2 Solderability (see 4.6.2.3). Terminal solderability shall meet the temperature and time requirements as specified in MIL-STD-883, method 2003, or as specified (see 3.1).

3.6 Maintainability. The transmitter shall require no preventive maintenance.

3.7 Transportability. The transmitter operational performance specified herein shall not be affected following prolonged storage or shipment under environmental conditions for shipment and storage specified herein, and as required for packaging for shipment (see 5.1).

3.8 Performance requirements. The transmitter shall meet all performance requirements of 3.8.1 and 3.8.2 when an electrical signal with characteristics as specified (see 3.8.3.1) is input to the transmitter.

3.8.1 Optical output signal properties.

3.8.1.1 Peak emission wavelength (see 4.6.3.2.1 and 6.6.12). The peak emission wavelength shall be as specified (see 3.1).

3.8.1.2 Spectral width (see 4.6.3.2.2 and 6.6.14). The spectral width shall be as specified (see 3.1).

3.8.1.3 Optical power levels (see 4.6.3.2.3). The minimum and maximum optical power output levels shall be as specified (see 3.1).

3.8.2 Electro-optical properties.

3.8.2.1 Signal-to-noise ratio (SNR) (see 4.6.3.3.1 and 6.6.13). The minimum output SNR shall be as specified (see 3.1), over the specified signal frequency range.

3.8.2.2 Frequency range (see 4.6.3.3.2 and 6.6.6). The transmitter shall convert the electrical input signals into optical signals over the frequency range as specified (see 3.1).

3.8.2.3 Total harmonic distortion (see 4.6.3.3.3 and 6.6.15). The total harmonic distortion shall be as specified (see 3.1).

3.8.2.4 Intermodulation distortion (see 4.6.3.3.4 and 6.6.9). The intermodulation distortion shall be as specified (see 3.1).

3.8.2.5 Amplitude distortion (see 4.6.3.3.5 and 6.6.1). The amplitude distortion shall be as specified (see 3.1).

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3.8.2.6 Group delay (see 4.6.3.3.6 and 6.6.7). The group delay shall be as specified (see 3.1).

3.8.3 Electrical input signal properties.

3.8.3.1 Analog input signal (see 4.6.3.1.2 and 6.6.2). The electrical input signal interface requirements for input impedance, signal levels, and modulation waveform shall be as specified (see 3.1).

3.8.3.2 Short circuit protection (see 4.6.3.4). The transmitter shall not experience any degradation of performance following application and removal of an electrical short between any required functional pin and logic ground or between any functional pin and supply voltage.

3.8.4 Power supply voltage and current (see 4.6.3.5). The transmitter shall operate with one or more of the following nominal direct current (dc) supply voltages: plus and/or minus 5 volts dc, plus and/or minus 12 volts dc, plus and/or minus 18 volts dc, plus and/or minus 24 volts dc, or plus and/or minus 28 volts dc as specified (see 3.1). The transmitter shall meet all electrical and optical performance requirements specified herein while operating with specified power supply voltages, currents, and their tolerances (see 3.1). The total transmitter power consumption shall not exceed the level specified (see 3.1).

3.9 Safety. The handling, installation, and operation of the transmitter shall not endanger personnel by direct contact or by long-term effects due to deterioration of any component within the transmitter construction. Transmitter design and operation shall conform to the safety standards of MIL-STD-454, requirement 1.

3.9.1 Laser safety. When laser transmitters are specified (see 1.2.1), the transmitters shall be labeled as specified (see 3.1).

3.10 Environmental requirements. The transmitter shall meet all performance requirements under the following environmental conditions and as specified (see 3.1).

3.10.1 Temperature extremes (see 4.6.4.1). The transmitter shall meet the performance requirements of 3.8.1 and 3.8.2 at the operating temperature extremes. The transmitter temperature range shall be as specified (see 3.1), in accordance with the ranges in table I. The change in optical output power over the temperature range shall be less than 4.0 decibels (dB), or as specified (see 3.1). The transmitter shall have no physical damage (see 4.6.2.4).

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TABLE I. Transmitter temperature ranges.

Temperature range	Operating (°C)	Nonoperating (°C)
1	0 to +85	-62 to +85
2	-28 to +85	-62 to +85
3	-54 to +85	-62 to +85
4	0 to +125	-62 to +125
5	-28 to +125	-62 to +125
6	-54 to +125	-62 to +125

3.10.2 Vibration (see 4.6.4.2). The transmitter shall meet the performance requirements of 3.8.1.2 and 3.8.2, and shall have no cracks, chips, warping, bending, or other physical damage (see 4.6.2.4). The transmitter shall remain hermetically sealed (see 4.6.2.1).

3.10.3 Thermal shock (see 4.6.4.3). Over the nonoperating temperature range specified (see 3.10.1), the transmitter shall meet the performance requirements of 3.8.1.2 and 3.8.2, and shall have no cracks, breaks, or other physical damage (see 4.6.2.4). The transmitter shall remain hermetically sealed (see 4.6.2.1).

3.10.4 Impact shock (see 4.6.4.4). The transmitter shall meet the performance requirements of 3.8.1.2 and 3.8.2, and shall have no cracks, breaks, or other physical damage (see 4.6.2.4). The transmitter shall remain hermetically sealed (see 4.6.2.1).

3.10.5 Salt spray (see 4.6.4.5). The transmitter percent corrosion shall be in accordance with MIL-STD-883, method 1009, test condition C.

3.10.6 Explosive atmosphere (see 4.6.4.6). The transmitter shall operate safely in flammable atmospheres without causing an explosion.

3.10.7 Fungus resistance (see 4.6.4.7). All materials used in the construction of the transmitter shall meet the requirements of MIL-STD-454, requirement 4. Materials not identified in MIL-STD-454, requirement 4, as fungus inert, shall be in accordance with grade 1 of MIL-STD-810, method 508.

3.10.8 Environmental screening (see 4.3.3 and 4.3.4). The transmitter shall be environmentally screened by random vibration testing and by temperature cycling.

3.11 Mechanical requirements (see 4.6.5).

3.11.1 Cable pull-out force (see 4.6.5.1). The minimum cable-to-transmitter pull-out strength shall be a minimum of 50 percent of the specified tensile strength of the attached cable or 100 newtons, whichever is less. There shall be

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no cable jacket damage, cable clamp failure, cable-to-transmitter seal damage, distortion or bending of transmitter parts, or cable disengagement from the clamp (see 4.6.3.4). The transmitter shall meet the requirements of 3.8.1.2.

3.11.2 Twist (see 4.6.5.2). Cable-to-transmitter seals shall not be damaged nor shall any other physical damage occur (see 4.6.2.4). The transmitter shall meet the requirements of 3.8.1.2.

3.11.3 Connector pull-out force (see 4.6.5.3). The minimum connector-to-transmitter pull-out strength shall be one-half of the specified tensile strength of the attached connector or 100 newtons, whichever is less. There shall be no connector-to-transmitter seal damage, distortion or bending of transmitter parts, or connector disengagement from the transmitter (see 4.6.2.4). The transmitter shall meet the requirements of 3.8.1.2.

3.11.4 Fiber dynamic strength (see 4.6.5.4). The minimum fiber dynamic strength shall be 50 percent of the specified proof strength of the attached fiber. There shall be no fiber-to-transmitter seal damage, or distortion or bending of transmitter parts (see 4.6.2.4). The transmitter shall meet the requirements of 3.8.1.2.

3.12 Nuclear radiation resistance. Unless otherwise specified (see 3.1), the performance requirements of 3.8 shall be met when tested in accordance with 4.6.6.

3.13 Electromagnetic effects (see 4.6.7). The transmitter shall meet the conducted and radiated emission requirements and the conducted and radiated susceptibility requirements of MIL-STD-461, part 5, as specified (see 3.1). Above-deck or below-deck application shall be as specified (see 3.1). The transmitter shall meet the requirements of 3.8.1.2 and 3.8.2.

3.14 Electrostatic discharge sensitivity (ESDS) (see 4.6.8). The transmitter shall meet the requirements of 3.8, and the requirements for ESDS class 2 devices in MIL-STD-883.

3.15 Dimensions (see 4.6.9). The transmitter shall be enclosed in a physical envelope with detailed dimensions as specified (see 3.1).

3.16 External finish (see 4.6.10). The transmitter case finish shall comply with the package finish requirements of MIL-M-38510. The finish shall maintain legibility of the unit part number and required markings (see 3.18). Unless otherwise specified (see 3.1), the pin finish shall comply with MIL-M-38510, finish type B.

3.17 Identification marking (see 4.6.11). Unless otherwise specified (see 3.1), the transmitter shall be marked in accordance with MIL-M-38510. The assigned part number shall use the numbering schemes specified (see 6.8).

3.18 Workmanship. Transmitter design shall meet all physical and interconnection requirements of this specification. There shall be no damaged parts, loose pins, cracked or abused surfaces, illegible markings, or inconsistencies in

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finish or appearance. The transmitter shall meet the requirements herein for visual inspection prior to final capping of the transmitter case and for visual inspection of the finished product (see 4.3.1).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Screening (see 4.3).
(b) First article inspection (see 4.4).
(c) Quality conformance inspection (see 4.5).

4.3 Screening. Each transmitter to be delivered or submitted for inspection shall have been subjected to and passed all of the following screening tests in the order shown (see table II). Transmitters with cable pigtails may be subjected to these tests before the pigtail exterior is installed over the coated fiber. For those transmitters with pigtails to be cabled after screening, a temporary strain relief may be installed at the transmitter-pigtail interface for the screening tests.

TABLE II. Screening inspection.

Inspection	Requirement	Test
Internal visual	3.18	4.3.1
Burn-in	3.3.1	4.3.2
Random vibration	3.10.8	4.3.3
Temperature cycling	3.10.8	4.3.4
External visual	3.18	4.6.2.4
Hermetic seal	3.5.4	4.6.2.1

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4.3.1 Internal visual (see 3.18). Internal visual inspection (pre-cap) shall be performed in accordance with MIL-STD-883, method 2017.

4.3.2 Burn-in (see 3.3.1). Burn-in testing shall be in accordance with MIL-STD-883, method 1015. Unless otherwise specified (see 3.1), all samples shall be tested for a minimum of 160 hours at the maximum specified temperature (see 3.10.1). Maximum power supply voltage and the appropriate signal voltage to provide maximum optical power level shall be applied to the transmitter during the test (see 3.8.4 and 3.8.1.3).

4.3.3 Random vibration (see 3.10.8). Random vibration testing shall be in accordance with MIL-STD-883, method 2026, condition II, test condition C. No power shall be applied during this test.

4.3.4 Temperature cycling (see 3.10.8). Temperature cycling shall be in accordance with MIL-STD-883, method 1010, using the specified nonoperating temperature range in table I. No power shall be applied during this test.

4.4 First article inspection (see 3.2). Inspection shall be in accordance with table III. The units submitted for testing shall be representative of the production process. The sample units used for groups II, III, and IV inspections shall have been subjected to and meet the requirements of the screening inspections (see 4.3) and group I inspections.

TABLE III. First article inspection.

Inspection	Requirement	Test	Sample size
Group I			
Identification marking	3.17	4.6.11	see 4.4.1
Peak emission wavelength	3.8.1.1	4.6.3.2.1	1/
Spectral width	3.8.1.2	4.6.3.2.2	1/
Optical power levels	3.8.1.3	4.6.3.2.3	1/
Signal to noise ratio	3.8.2.1	4.6.3.3.1	1/
Frequency range	3.8.2.2	4.6.3.3.2	1/
Total harmonic distortion	3.8.2.3	4.6.3.3.3	1/
Intermodulation distortion	3.8.2.4	4.6.3.3.4	1/
Amplitude distortion	3.8.2.5	4.6.3.3.5	1/
Group delay	3.8.2.6	4.6.3.3.6	1/
Analog input signal	3.8.3.1	4.6.3.1.2	1/

See footnotes at end of table.

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TABLE III. First article inspection - Continued.

Inspection	Requirement	Test	Sample size
Short circuit protection	3.8.3.2	4.6.3.4	1/
Power supply voltage and current	3.8.4	4.6.3.5	1/
Dimensions	3.15	4.6.9	1/
External finish	3.16	4.6.10	1/
Workmanship	3.18	4.6.2.4	1/
Temperature extremes	3.10.1	4.6.4.1	1/
Group II			
Terminal strength	3.5.5.1	4.6.2.2	1
Solderability	3.5.5.2	4.6.2.3	2/
Thermal shock	3.10.3	4.6.4.3	1
Impact shock	3.10.4	4.6.4.4	3/
Vibration	3.10.2	4.6.4.2	3/
Salt spray	3.10.5	4.6.4.5	3/
Hermetic seal	3.5.4	4.6.2.1	3/
Mechanical tests	3.11	4.6.5	3/
Explosive atmosphere	3.10.6	4.6.4.6	3/
Fungus resistance	3.10.7	4.6.4.7	3/
Group III			
Electromagnetic effects	3.13	4.6.7	1
Electrostatic discharge sensitivity	3.14	4.6.8	4/
Group IV			
Reliability	3.3	4.6.1	see 4.3.3
Nuclear radiation resistance	3.12	4.6.6	4

See footnotes at top of next page.

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1/ The same samples shall be used for all group I inspections.
2/ The same sample shall be used as in the terminal strength inspection.
3/ The same sample shall be used as in the thermal shock inspection.
4/ The same sample shall be used as in the electromagnetic effects inspection.

4.4.1 Samples. Unless otherwise specified (see 6.4), the number of sample units to be subjected to first article inspection shall be seven transmitters, not including those required for reliability testing (see 3.3). For transmitters with cable pigtails, those units to be subjected to group II tests may have the pigtail exterior installed after the hermetic seal test. For those transmitters with pigtails to be cabled after hermetic seal testing, a temporary strain relief may be installed at the transmitter/pigtail interface for those tests prior to hermetic seal testing.

4.4.2 Failures. One or more failures shall be sufficient cause for refusal to grant first article approval. When first article test units are taken from a larger lot and the first article is disapproved, none of the units from that lot shall be delivered.

4.5 Quality conformance inspection. Quality conformance inspection shall include the inspections of groups A, B, and C as specified in tables IV, V, and VI (see 6.5). Each transmitter shall successfully pass all screening tests (see 4.3) prior to performing group A inspections.

4.5.1 Group A inspection. The group A inspections shall be nondestructive tests, consisting of visual, mechanical and performance tests as provided in table IV and as specified (see 3.1). Each transmitter shall comply with the group A requirements prior to delivery and prior to groups B and C testing, and shall be tested in the order shown.

TABLE IV. Group A inspection.

Inspection	Requirement	Test
Identification marking	3.17	4.6.11
Peak emission wavelength	3.8.1.1	4.6.3.2.1
Spectral width	3.8.1.2	4.6.3.2.2
Optical power levels	3.8.1.3	4.6.3.2.3
Signal-to-noise ratio	3.8.2.1	4.6.3.3.1
Frequency range	3.8.2.2	4.6.3.3.2
Total harmonic distortion	3.8.2.3	4.6.3.3.3
Intermodulation distortion	3.8.2.4	4.6.3.3.4
Amplitude distortion	3.8.2.5	4.6.3.3.5
Group delay	3.8.2.6	4.6.3.3.6

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TABLE IV. Group A inspection - Continued.

Inspection	Requirement	Test
Analog input signal	3.8.3.1	4.6.3.1.2
Short circuit protection	3.8.3.2	4.6.3.4
Power supply voltage and current	3.8.4	4.6.3.5
Temperature extremes	3.10.1	4.6.4.1

4.5.2 Group B inspection. The group B inspections shall be nondestructive tests in table V and as specified (see 3.1).

4.5.2.1 Inspection sequence. For transmitters with cable pigtails, those units to be subjected to group B inspection may have the pigtail exterior installed after the hermetic seal test. For those transmitters with pigtails to be cabled after hermetic seal testing, a temporary strain relief may be installed at the transmitter interface for all tests prior to hermetic seal testing. Group B inspections shall be tested in the sequence shown.

TABLE V. Group B inspection.

Inspection	Requirement	Test
Impact shock	3.10.4	4.6.4.4
Thermal shock	3.10.3	4.6.4.3
Hermetic seal	3.5.4	4.6.2.1
Mechanical tests	3.11	4.6.5

4.5.3 Group C inspection. The group C inspections shall consist of tests in table VI and as specified (see 3.1). These tests may be destructive to sample parts. Unless otherwise specified (see 6.5), samples for each test shall consist of subsets of group B test samples. For transmitters with cable pigtails, the cable exterior may be removed prior to hermetic seal testing.

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TABLE VI. Group C inspection.

Inspection	Requirement	Test
Reliability	3.3	4.6.1
Nuclear radiation resistance	3.12	4.6.6
Terminal strength	3.5.5.1	4.6.2.2
Electromagnetic effects	3.13	4.6.7
Electrostatic discharge sensitivity	3.14	4.6.8
Vibration	3.10.2	4.6.4.2
Hermetic seal	3.5.4	4.6.2.1
Explosive atmosphere	3.10.6	4.6.4.6
Solderability	3.5.5.2	4.6.2.3
Salt spray	3.10.5	4.6.4.5
Fungus resistance	3.10.7	4.6.4.7

4.6 Methods of inspection.

4.6.1 Reliability (see 3.3). The transmitter failure rate shall be determined for class B devices (as defined in MIL-STD-883) using MIL-STD-883, method 1016, except that the minimum ambient temperature shall be 150 degrees Celsius (°C). Optical power measurements shall be made using an input electrical signal corresponding to the characteristics specified in 3.8.3.1 at the maximum optical output power level and maximum specified frequency. The power supply shall be set to the maximum specified magnitude (see 3.8.4). Proper device operation shall be determined by verifying the average optical output power using the test specified in 4.6.3.2.3. The number of test units shall be as specified (see 6.2).

4.6.2 Design and construction inspections.

4.6.2.1 Hermetic seal (see 3.5.4). The transmitter shall be tested in accordance with MIL-STD-883, method 1014, methods A1 and C. For transmitters with cable pigtails see 4.3, 4.4.1, 4.5.2.1, and 4.5.3.

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4.6.2.2 Terminal strength (see 3.5.5.1). The transmitter electrical terminals shall be tested in accordance with MIL-STD-883, method 2004. Test condition A shall be used for tension; condition B1 shall be used for stress; and condition B2 shall be used for electrical pin fatigue.

4.6.2.3 Solderability (see 3.5.5.2). The transmitter electrical pins shall be tested in accordance with MIL-STD-883, method 2003.

4.6.2.4 Visual inspection (see 3.18). Transmitters shall be visually inspected in accordance with MIL-STD-883, method 2009.

4.6.3 Performance tests.

4.6.3.1 General. The performance tests for the transmitter shall be performed with the transmitter having not less than 1-meter length of fiber connected. If used, the connector and the fiber or cable comprising the pigtail shall meet the requirements as specified (see 3.1). Measurements shall be made at the output end of the optical pigtail while applying the appropriate cladding mode stripper (see 6.6.3). For single-mode fiber, the pigtail shall be wrapped one turn around a 30-millimeter diameter mandrel.

4.6.3.1.1 Standard test conditions. Unless otherwise specified (see 3.1), the following ambient test conditions shall be used:

(a) Temperature: 25 plus 3, minus 5°C.
(b) Relative humidity: 50 plus or minus 30 percent.
(c) Atmospheric pressure: 725 plus or minus 75 millimeters (mm) of mercury.

4.6.3.1.2 Analog input signal (see 3.8.3.1). The transmitter performance shall be verified with the specified electrical analog input signal using the procedures of 4.6.3.2 and 4.6.3.3.

4.6.3.2 Optical output properties (see 3.8.1). Unless otherwise specified (see 3.1), the optical performance parameters shall be verified using the test procedures specified herein and in accordance with the following:

(a) The transmitter shall be connected to the apparatus.
(b) The power supply voltage shall be set to the minimum magnitude specified (see 3.8.4).
(c) The test instrument parameters shall have the capability of measuring the signal under test with a tolerance of 3 percent over the range specified for the device.
(d) The optical power meter shall operate over the specified wavelength range and shall have a dynamic range suitable for the specified optical power measurements. The equipment shall provide repeatable readings with less than 3 percent variation.
(e) A suitable analog input signal with the applicable modulation format and signal amplitude as specified (see 3.1), shall be applied to the input of the transmitter under test.
(f) The results shall be verified in accordance with the requirements of 3.8.1.

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(g) The power supply shall be set to the maximum magnitude specified (see 3.8.4). Steps (c) through (f) shall be repeated.
(h) The power supply voltage shall be set to the specified nominal magnitude (see 3.8.4). Steps (c) through (f) shall be repeated.

4.6.3.2.1 Peak emission wavelength (see 3.8.1.1). In accordance with 4.6.3.1, 4.6.3.2, and figures 1 and 2, the peak emission wavelength shall be determined at reference point A on figure 2, by observing point D.

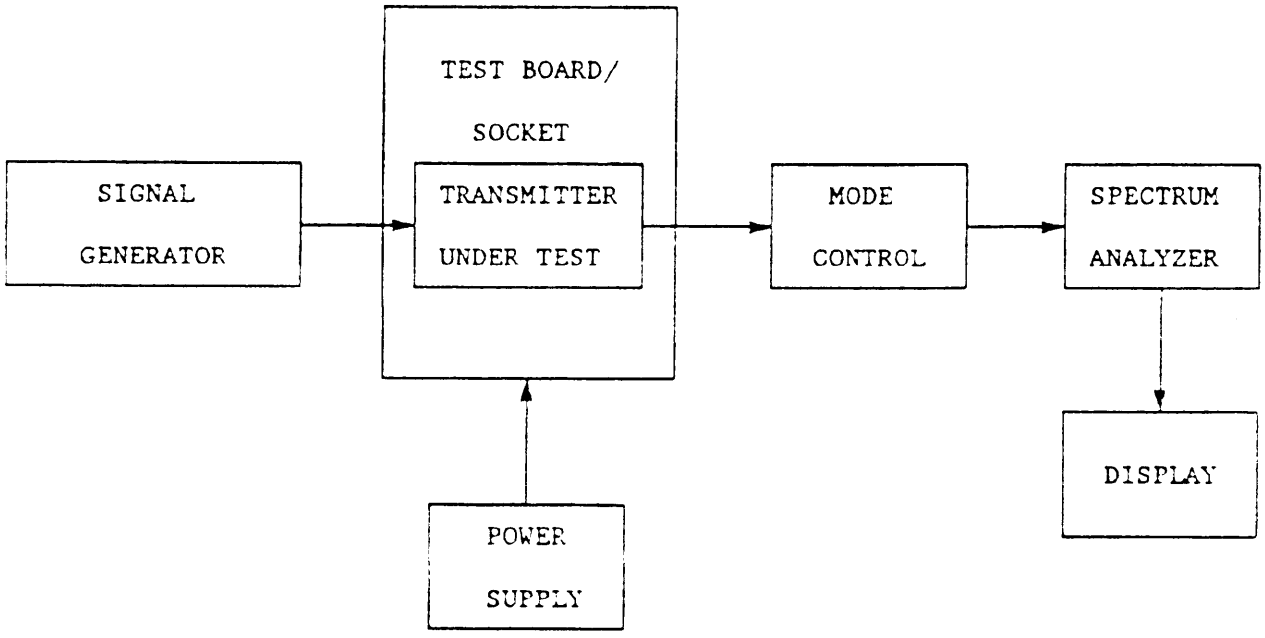
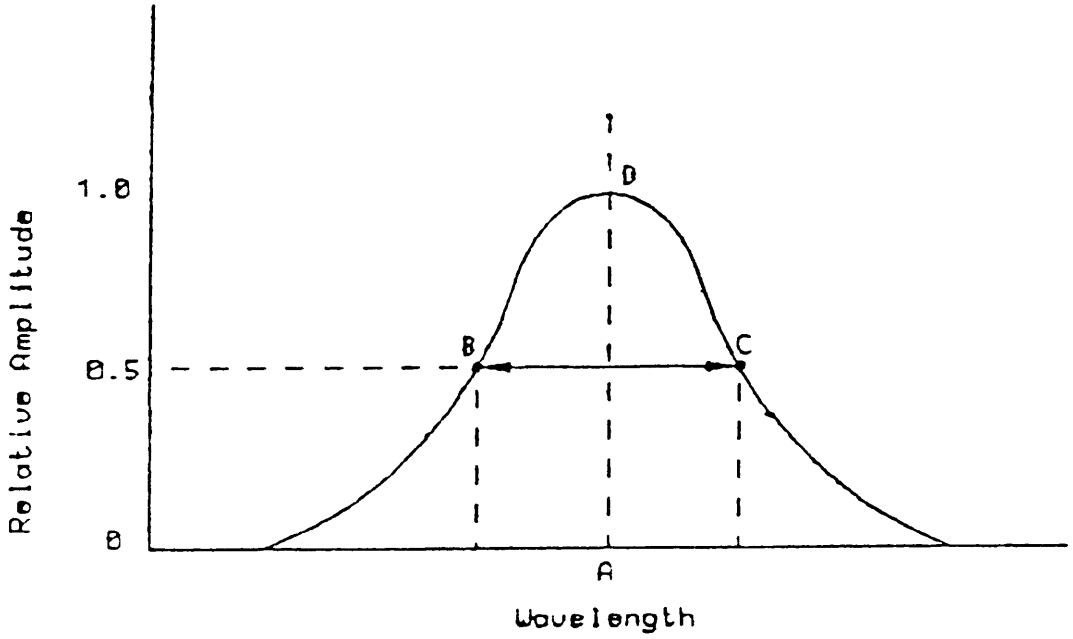


FIGURE 1. Spectral performance test set-up.

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A - Peak emission wavelength
BC - Full-width half-maximum
D - Peak emission power level

FIGURE 2. Optical spectral envelope (LEDs).

4.6.3.2.2 Spectral width (see 3.8.1.2). In accordance with 4.6.3.1, 4.6.3.2, and figures 1 through 3, the spectral width shall be measured as follows:

(a) For LED transmitters measure the spectral width at the full-width half-maximum (FWHM) power points as shown between points B and C on figure 2.

(b) For multifrequency laser transmitters, calculate the root mean square (rms) width, as shown on figure 3 between points A and B. The RMS width shall be calculated as follows: RMS = $\Delta\lambda = [\ 1/P_o\ \Sigma_i\ \ P_i\ (\lambda_i - \lambda_c)^2\ ]^{0.5}$ where P_o = the sum of the peak optical output powers (see 4.6.4.3); P_i = peak optical power at each output wavelength; λ_i = center of an individual optical output wavelength at peak power P_i; λ_c = statistical center wavelength given by $\lambda_c = 1/P_o\ \Sigma_i\ P_i\ \lambda_i$.

(c) For single frequency laser transmitters measure the full width at minus 10 dB referenced to 0 dB at the peak emission power level.

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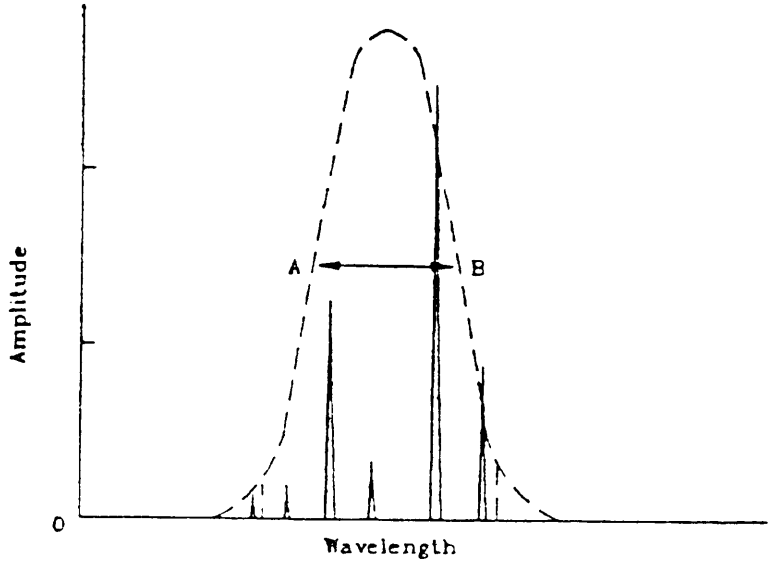


FIGURE 3. Optical spectral envelope (multifrequency lasers).

4.6.3.2.3 Optical power levels (see 3.8.1.3). The optical power output levels shall be measured in accordance with 4.6.3.1, 4.6.3.2, and the test set-up shown on figure 4.

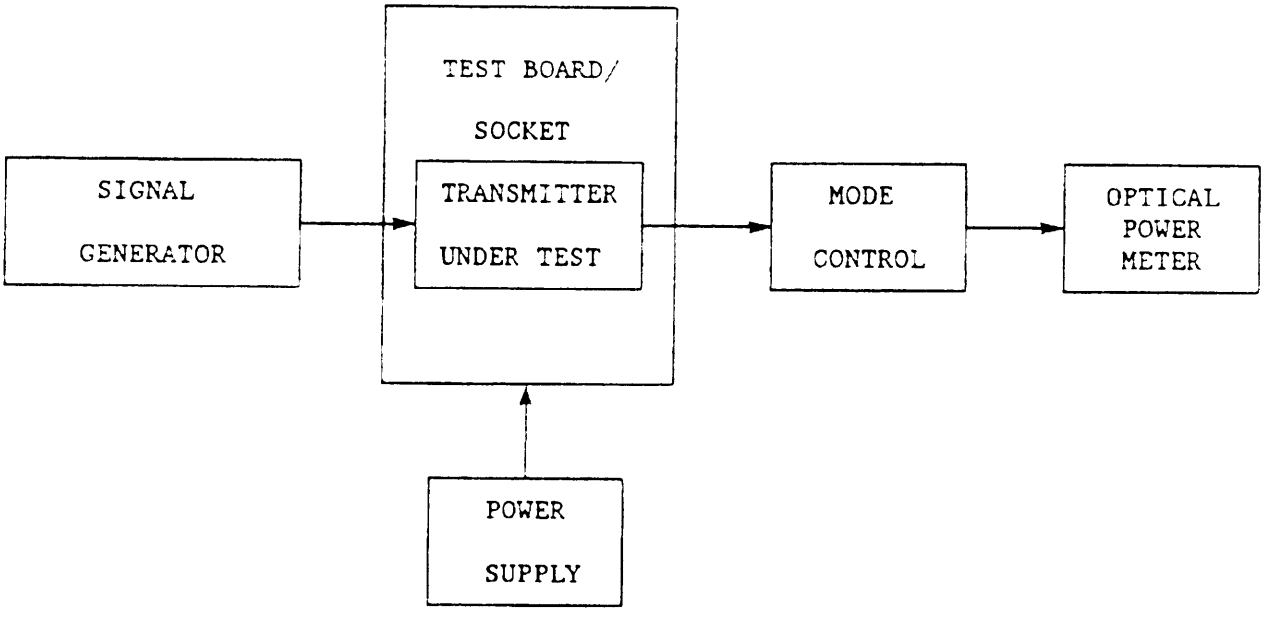


FIGURE 4. Optical power level test set-up.

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4.6.3.3 Electro-optical properties (see 3.8.2). Unless otherwise specified (see 3.1), the electro-optical characteristics shall be verified using the test procedures specified in 4.6.3.3.1 through 4.6.3.3.6 and in accordance with the following:

(a) The transmitter shall be connected to the apparatus.
(b) The power supply voltage shall be set to the minimum magnitude specified (see 3.8.4).
(c) The test instrument parameters shall have the capability of measuring the signal under test with a tolerance of 3 percent over the range specified for the device.
(d) A suitable analog input signal of signal amplitude and maximum frequency, as specified herein, and with the applicable modulation format, shall be applied to the input of the transmitter under test.
(e) The analog input signal shall be applied via a passband filter as specified in the applicable transmitter specification sheet (see 3.1).
(f) The results shall be verified in accordance with the requirements of 3.8.2.
(g) The power supply shall be set to the maximum magnitude specified (see 3.8.4). Steps (c) through (f) shall be repeated.
(h) The power supply voltage shall be set to the specified nominal magnitude (see 3.8.4). Steps (d) through (f) shall be repeated.

4.6.3.3.1 Signal-to-noise ratio (SNR) (see 3.8.2.1). SNR shall be verified in accordance with 4.6.3.1, 4.6.3.3, and figure 5 over the frequency range specified (see 3.8.2.2).

4.6.3.3.2 Frequency range (see 3.8.2.2). In accordance with 4.6.3.1, 4.6.3.3, and the test set-up specified on figure 5, the signal generator shall inject a signal of constant amplitude into the transmitter. The signal shall be varied over the frequency range specified (see 3.8.2.2). The spectrum analysis shall be determined from the recovered signal that shall be displayed at the spectrum analyzer. A receiver complementary to the transmitter under test shall be used.

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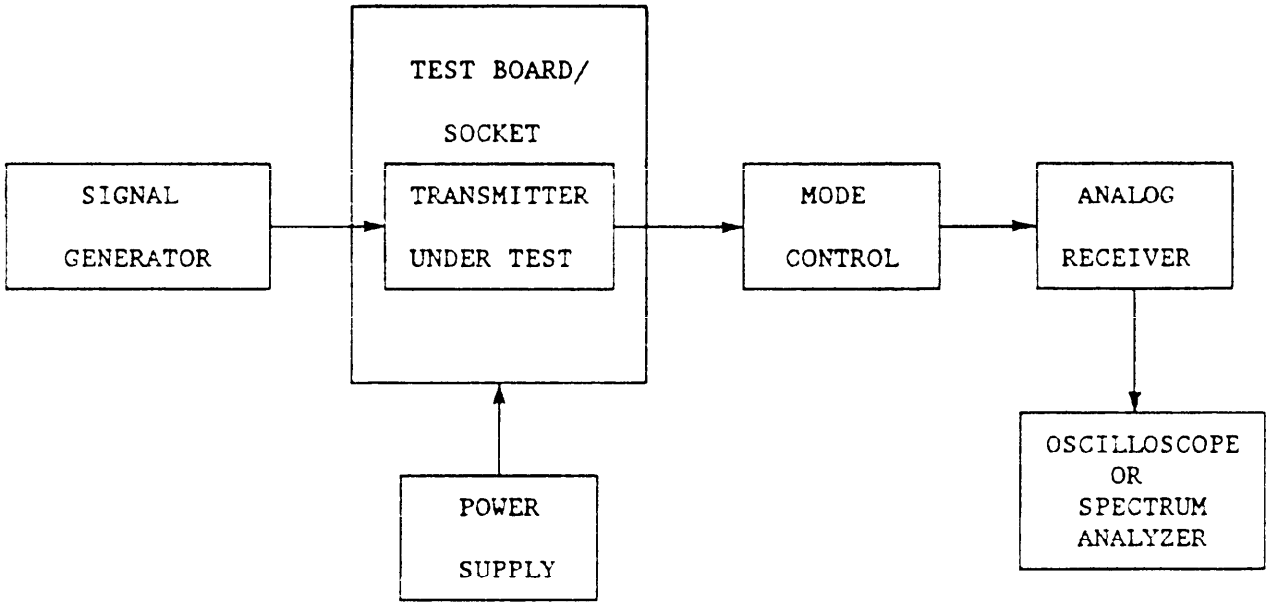


FIGURE 5. Electro-optical characteristics test set-up.

4.6.3.3.3 Total harmonic distortion (THD). In accordance with 4.6.3.1, 4.6.3.3, and the test set-up specified on figure 5, a single frequency signal with a spectral purity better than the THD specified (see 3.8.2.3) shall be injected into the transmitter under test. The THD shall be determined from the spectral analysis obtained from the output of the complementary receiver.

(a) The input signal should be of the form

$$x(t) = A_o \sin \omega_o t$$

(b) For an output of the form

$$y(t) = a_o + a_1\cos(\omega_c t + \phi_1) + a_2\cos(2\omega_o t + \phi_2) + a_3\cos(3\omega_c t + \phi_3) + \ldots$$

the percent THD shall be equal to $([\sqrt{\Sigma^{m}_{n-2}\ a_n^2}\ ]\ /a_1\ )$ X 100

where a_n is the peak value of the output at nf_c (where f_c is the frequency associated with the input ω_o). The last frequency to be measured, mf_o, shall be as specified (see 3.1).

(c) The THD shall be measured at the test input frequencies specified (see 3.1) to provide measurement of the THD over the transmitter frequency range specified (see 3.8.2.2).

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4.6.3.3.4 Intermodulation distortion (see 3.8.2.4). In accordance with 4.6.3.1, 4.6.3.3, and the test set-up specified on figure 5, two constant amplitude signals shall be swept in frequency and injected into the transmitter under test. The two signals shall be selected in accordance with the following:

(a) The input shall be of the form

$$x(t) = A \sin \omega_1 t + A \sin \omega_2 t$$

where A sin $\omega_1 t$ is one of two input signals and A sin $\omega_2 t$ is the other input signal. The frequencies f_1 and f_2 associated with ω_1 and ω_2 shall be as specified (see 3.1). Multiple sets of f_1 and f_2 as specified (see 3.1) shall be used for test inputs to provide measurement of the IMD over the transmitter frequency range specified (see 3.8.2.2).

(b) Unless otherwise specified, the difference between frequencies f_1 and f_2 shall be not greater than 2 percent of the frequency range specified (see 3.8.2.2).

The percent of IMD shall be measured for any third-order cross products whose frequencies occur within the specified receiver frequency range (see 3.8.2.2) and that are associated with the following:

(a) The output of the receiver shall be of the form

$$y(t) = a_1 x(t) + a_2 x^2(t) + a_3 x^3(t)$$

where $a_1 x(t)$ is the first-order output term and $a_3 x^3(t)$ is the third-order output term.

(b) The amplitude of the third order cross products that are part of the third-order output term, to be considered for measurement, shall be at the frequencies $2f_1 + f_2$, $2f_2 + f_1$, $2f_1 - f_2$, and $2f_2 - f_1$.

(c) The percent of distortion at each applicable cross product frequency shall be 100 times the amplitude at the cross product frequency divided by the amplitude of the first-order term a_1.

4.6.3.3.5 Amplitude distortion (see 3.8.2.5). In accordance with 4.6.3.1, 4.6.3.3, and the test set-up specified on figure 5, apply a sinusoidal input signal of frequency and amplitude as specified (see 3.8.2.2) and sweep the signal source over the frequency range as specified (see 3.1). Measure the output signal amplitude with respect to frequency and determine the amplitude distribution in decibels.

4.6.3.3.6 Group delay (see 3.8.2.6). The test set-up shall be in accordance with 4.6.3.1 and 4.6.3.3. A narrow-band, amplitude modulated, sinusoidal signal, as specified (see 3.1), shall be injected into the transmitter under test. The sinusoidal signal shall be varied in frequency as specified (see 3.1). The transmitter shall be coupled to a complementary analog receiver via a pigtail assembly subjected to mode control. An oscilloscope shall be used to display the recovered signal output from the receiver. Group delay shall be measured as the difference in phase of the envelope over the frequency range tested.

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4.6.3.4 Short circuit protection (see 3.8.3.2). The short circuit protection shall be tested in accordance with MIL-STD-883, method 3011.

4.6.3.5 Power supply voltage and current. The maximum current shall be measured and verified in accordance with MIL-STD-883, method 3005, using the nominal supply voltage specified (see 3.8.4).

4.6.4 Environmental tests (see 3.10). The following environmental tests shall be performed as specified in the referenced test methods.

4.6.4.1 Temperature extremes. The performance of the transmitter shall be verified in accordance with 4.6.3 at both of the operating temperature extremes as specified (see 3.10.1). At each extreme the transmitter shall have attained thermal equilibrium for at least 30 minutes before any measurements are made.

4.6.4.2 Vibration (see 3.10.2). The transmitter shall be tested for effects of vibration in accordance with MIL-STD-883, method 2026, test condition II, test condition C, and method 2007 test condition A. Verification of performance shall be in accordance with 4.6.3.2.3 and 4.6.3.3 during and after the test.

4.6.4.3 Thermal shock (see 3.10.3). The effects of thermal shock on transmitter performance shall be tested using MIL-STD-883, method 1011, over the nonoperating temperature range specified (see 3.10.1). Verification of performance shall be in accordance with 4.6.3.2.3 and 4.6.3.3 after the test.

4.6.4.4 Impact shock (see 3.10.4). The transmitter shall be subjected to physical shock using MIL-STD-883, method 2002, test condition B. Unless otherwise specified (see 3.1), the optical power level shall be monitored during this test with equipment having a time resolution of 1 μs. Verification of performance shall be in accordance with 4.6.3.2.3 during the test, and 4.6.3.2.3 and 4.6.3.3 after the test.

4.6.4.5 Salt spray (see 3.10.5). The transmitter shall be tested for effects of a salt atmosphere in accordance with MIL-STD-883, method 1009, test condition C.

4.6.4.6 Explosive atmosphere (see 3.10.6). The transmitter safe operation within explosive atmospheric environments shall be tested using MIL-STD-202, method 109, under conditions as specified (see 3.1).

4.6.4.7 Fungus resistance (see 3.10.7). Transmitters and materials used in the construction of the transmitters that are not identified as fungus inert shall be tested in accordance with MIL-STD-810, method 508.

4.6.5 Mechanical tests (see 3.11).

4.6.5.1 Cable pull-out force (see 3.11.1). The transmitter shall be tested in accordance with EIA 455-6. The axial tensile load shall be applied and maintained for 10 minutes. The output optical power level shall be measured during and after the test (see 4.6.3.2.3).

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4.6.5.2 Twist (see 3.11.2). The transmitter shall be tested in accordance with EIA 455-36, with the exception that the test sample shall be a properly pigtailed transmitter. The number of loads to be applied shall be one. The output optical power level shall be measured during and after the test (see 4.6.3.2.3).

4.6.5.3 Connector pull-out force (see 3.11.3). The transmitter shall be tested in accordance with EIA 455-6, with the exception that the test sample shall be a connector on the transmitter. The axial load shall be applied to and maintained for 10 minutes. The output optical power level shall be measured during and after the test (see 4.6.3.2.3).

4.6.5.4 Fiber dynamic strength (see 3.11.4). The fiber pigtail dynamic strength shall be tested as follows: The fiber pigtail shall have an axial tensile load applied at an angle of 45 degrees to the normal (see figure 6). With the load applied, the pigtail shall be rotated through one rotation (360 degrees). The output optical power level shall be measured during and after the test (see 4.6.3.2.3).

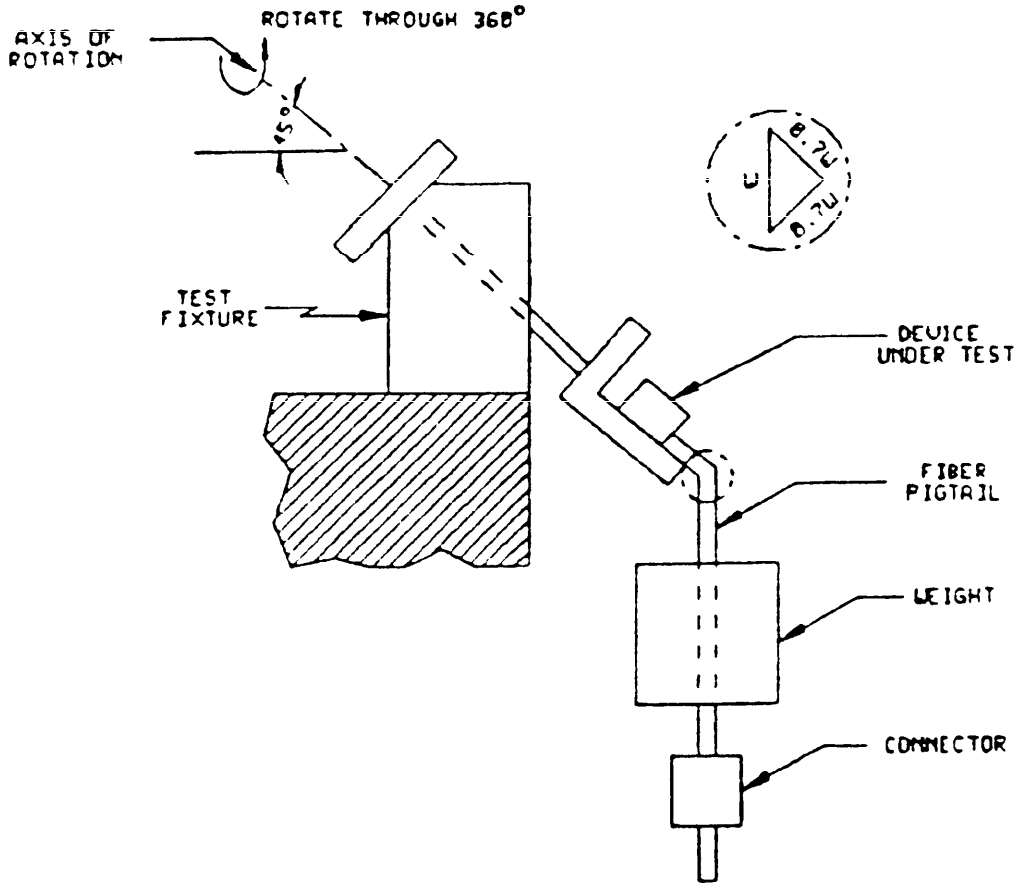


FIGURE 6. Fiber dynamic strength test set up.

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4.6.6 Nuclear radiation resistance (see 3.12). The transmitter shall be subjected to radiation in accordance with MIL-STD-883, methods 1017 (neutron) and 1019 (total dose). The optical power level shall be monitored continuously during the neutron and total dose tests. The dose rates and exposure times shall be as specified (see 6.2). Verification of performance shall be in accordance with 4.6.3.2.3 during the test and 4.6.3 after the test. The required test temperatures shall be the low and high operating temperatures, and 20°C.

4.6.7 Electromagnetic effects (see 3.13). The transmitter shall be tested for effects of electromagnetic emissions in accordance with MIL-STD-462 as applicable. Verification of performance shall be in accordance with 4.6.3.2.3 and 4.6.3.3 during and after the test.

4.6.8 Electrostatic discharge sensitivity (see 3.14). The transmitter shall be tested for sensitivity to electrostatic discharge in accordance with MIL-STD-883, method 3015. The electrostatic pulse shall not be less than 3000 volts or as specified (see 3.1). Verification of performance shall be in accordance with 4.6.3 after the test.

4.6.9 Dimensions (see 3.15). The transmitter shall be inspected for compliance with unit dimensions and tolerances using MIL-STD-883, method 2016.

4.6.10 External finish (see 3.16). The external finish of the transmitter case shall be verified using MIL-STD-883, method 2009.

4.6.11 Identification markings (see 3.17). The transmitter markings shall be examined for legibility and required information using MIL-STD-883, method 2009. The marking shall be legible and complete, and shall meet the resistance to solvents in accordance with MIL-STD-883, method 2015.

4.6.12 Inspection of packaging. Sample packages and packs and inspection of the preservation, packing, and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.7.)

5.1 Preservation and packing. The transmitter shall be packaged for shipment and storage in accordance with MIL-M-55565.

5.2 Marking. Marking on the transmitter packaging shall be in accordance with MIL-M-38510.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

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6.1 Intended use. The transmitters specified herein are intended for the generation of optically transmitted, digital data signals. The transmitter applications may include any signal data use within its operating range as specified in this document.

6.1.1 Engineering information. MIL-R-24737 is the companion specification for analog fiber optic shipboard receivers. Although the companion specification is not involved in the classification or testing of transmitters, it is often useful for evaluating the fiber optic link system.

6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification.
(b) Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
(c) Applicable specification sheet number, title, and date.
(d) When first article is required (see 3.2).
(e) When reliability testing is required, the number of test units required (see 3.3).
(f) The number of test units needed (see 4.6.1).
(g) Nuclear radiation dose rates and exposure times when this test is required (see 4.6.6).
(h) Part number of transmitter (see 6.8).
(i) Quantity required.

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DoD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
3.4	DI-E-2121	Certificate of compliance	----

The above DID's were those cleared as of the date of this specification. The current issue of DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.4 First article. When first article inspection is required, the items should be a first article sample. The first article should consist of 7 units. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for

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first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4.1 Lot size. The lot size may be specified contractually, if desired.

6.5 Quality conformance. Quality conformance inspections require contractual definition of the overall test program, including sample sizes and lot sizes, if appropriate (see 4.5).

6.6 Definitions.

6.6.1 Amplitude distortion. Amplitude distortion is distortion of a signal in a system, subsystem, or device when the output amplitude is not a linear function of the input amplitude under specified operating conditions.

6.6.2 Analog signal. An analog signal is a nominally continuous signal that is considered to be continuously variable and whose magnitude is made directly proportional to the data or to a suitable function of the data.

6.6.3 Cladding mode stripper. A cladding mode stripper is a material applied to cladding to allow electromagnetic energy being transmitted in the cladding to leave the cladding.

6.6.4 Fiber cladding. Fiber cladding is a transparent material that surrounds the core of an optical fiber and that has a lower index of refraction than the core material.

6.6.5 Fiber core. A fiber core is the central portion of an optical fiber. The core has a higher refractive index than the cladding that surrounds it. The bulk of light energy is confined to and propagates in the core. Single-mode fiber core diameters range from 2 to 11 μm, depending on the numerical aperture, core radius, and wavelength of the incident light.

6.6.6 Frequency range. Frequency range is the difference between the limiting frequencies of a continuous frequency band.

6.6.7 Group delay. Group delay is the distortion of a waveform made up of two or more different frequencies, caused by the difference in arrival time of each frequency at the output of a transmission system.

6.6.8 Interface. An interface is a concept involving the specification of the interconnection between two equipments of systems. The specification includes the type, quantity, and function of the interconnection circuits, and the type and form of signals to be interchanged by means of those circuits.

6.6.9 Intermodulation distortion. Intermodulation distortion is a nonlinear distortion characterized by the appearance of frequencies in the output, equal to the sums and differences of integral multiples of the component frequencies present in the input.

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6.6.10 Optical cable pigtail. An optical cable pigtail is a short length of fiber optic cable permanently fixed to a component and is used to couple optical power between it and another optical cable pigtail or fiber optic cable, such as a transmission cable.

6.6.11 Optical fiber pigtail. An optical fiber pigtail is a short length of optical fiber permanently fixed to a component and is used to couple optical power between it and another optical fiber pigtail or a transmission fiber.

6.6.12 Peak emission wavelength. The peak emission wavelength is the wavelength at which a light source emits its maximum optical power.

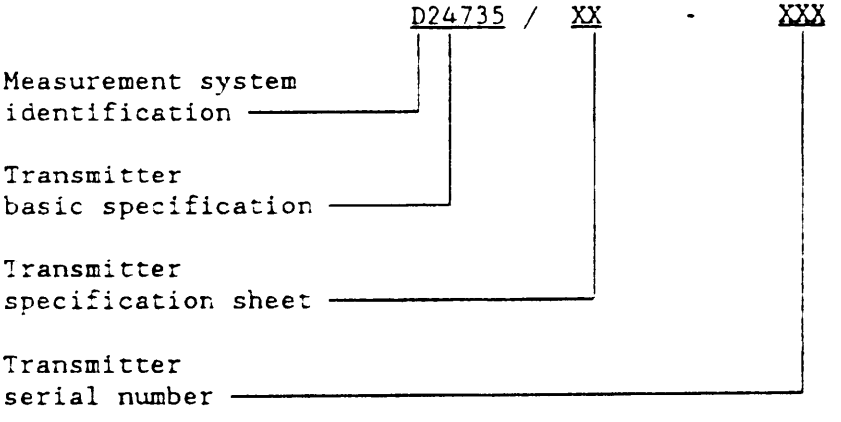
6.6.13 Signal-to-noise ratio (SNR). SNR is the ratio of the amplitude of the desired signal to the amplitude of noise signals.

6.6.14 Spectral width. Spectral width is the wavelength interval in which a radiated spectral quantity is a specified fraction of its maximum value. The fraction is usually taken at 0.5 of the maximum power level, and is the full-width half-maximum value.

6.6.15 Total harmonic distortion. Total harmonic distortion is the ratio of the sum of the powers of all harmonic frequency signals (other than the fundamental) to the power of the fundamental frequency signal at the output. This ratio is measured under specified conditions and is expressed in decibels.

6.7 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.8 Part or Identifying Number (PIN). The PINs to be used for transmitters acquired to this specification are created as follows:



Examples: D24735/01-001
D24735/01-002

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-24735(NAVY)	TRANSMITTERS, LIGHT SIGNAL, ANALOG, FI SHIPBOARD (METRIC), GENERAL SPECIFICATION FOR

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-24735(NAVY)	TRANSMITTERS, LIGHT SIGNAL, ANALOG, FIBER OPTIC, SHIPBOARD (METRIC), GENERAL SPECIFICATION FOR

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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